

Comparative Analysis of Classical Machine Learning and Transformer-Based Models for Bangla Sentiment Analysis on Social Media Data

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Abstract—Bangla sentiment analysis remains a challenging task due to the language’s morphological complexity and the scarcity of high-quality annotated datasets. This study presents a systematic evaluation of classical machine learning and transformer-based approaches on the Student-People Mass Uprising Public Sentiments Dataset, a corpus of 4,200 Bangla social media comments collected during the July Revolution in Bangladesh. We implement TF-IDF based models Logistic Regression, Naive Bayes, and Random Forest alongside fine-tuned transformer models including BanglaBERT, XLM-RoBERTa, and mBERT. Experimental results demonstrate that BanglaBERT achieves the highest performance with a Macro F1-score of 0.8726. However, a carefully tuned TF-IDF + Logistic Regression baseline achieves competitive performance with a Macro F1-score of 0.8445, highlighting the effectiveness of lightweight models on small datasets. The study provides insights into model selection for low-resource languages and emphasizes the importance of preprocessing strategies and feature representation in Bangla sentiment classification.

Index Terms—Bangla NLP, Sentiment Analysis, BanglaBERT, TF-IDF, Transformer Models, Low-Resource Languages, Social Media

I. INTRODUCTION

Natural Language Processing (NLP) for low-resource languages remains an active area of research, particularly for languages such as Bangla, which is spoken by over 265 million people worldwide. Despite its widespread usage, Bangla continues to face significant challenges in NLP due to its complex morphology, informal writing styles in social media, and limited availability of high-quality annotated datasets [1].

The July Revolution of 2024 in Bangladesh generated a large volume of user-generated content across social media platforms, reflecting diverse public opinions and sentiments. Analyzing such data is critical for understanding public perception, social dynamics, and policy implications. However, sentiment classification in this domain is non-trivial due to noise, domain-specific language, and platform-specific biases.

Traditional approaches to sentiment analysis rely on feature engineering techniques such as TF-IDF combined with classical machine learning models. More recently, transformer-based models have demonstrated strong performance by leveraging contextual embeddings. However, these models often require substantial data and computational resources, raising questions about their effectiveness on relatively small datasets.

This project aims to systematically evaluate and compare classical machine learning models and transformer-based approaches for Bangla sentiment classification. Using a curated dataset of social media comments related to the July Revolution, we investigate the trade-offs between model complexity and performance. The primary objective is to identify an effective and practical approach for sentiment analysis in low-resource Bangla settings while ensuring robustness and interpretability.

In low-resource settings such as Bangla, the availability of large annotated datasets is limited, making it essential to evaluate whether complex deep learning models consistently outperform simpler approaches. Understanding this trade-off is particularly important for practical deployment scenarios where computational resources may be constrained. Therefore, a comparative evaluation between classical machine learning models and transformer-based architectures provides valuable insight into model efficiency, scalability, and real-world applicability.

II. RELATED WORK

Bangla NLP has evolved significantly over the past decades, yet it remains constrained by limited resources and linguistic complexity. A comprehensive survey by Sen et al. [1] highlights the progression from classical machine learning methods to deep learning approaches, emphasizing the persistent challenges in Bangla text processing, including morphological variation and scarcity of annotated corpora.

Early work on Bangla sentiment analysis focused on character-level and word-level neural architectures. Haydar et al. [2] demonstrated that character-level recurrent neural networks outperform word-level models for Bangla text due to their ability to capture morphological patterns and handle noisy social media inputs. This finding established the importance of subword-level representations for Bangla sentiment classification.

The introduction of pretrained transformer models marked a major advancement in Bangla NLP. BanglaBERT, proposed by Bhattacharjee et al. [3], was trained on a large-scale Bangla corpus and demonstrated superior performance across multiple downstream tasks compared to multilingual models such as mBERT and XLM-RoBERTa. Its strong performance,

particularly in low-resource settings, makes it a suitable choice for sentiment analysis tasks involving limited data.

Subsequent research explored hybrid approaches combining multiple representations. Islam and Alam [4] introduced hybrid feature extraction techniques combining BanglaBERT embeddings with statistical methods, achieving improved performance on large-scale datasets. Similarly, Hoque et al. [5] demonstrated that ensemble transformer models outperform individual models in Bangla sentiment classification tasks.

The study most closely related to this work is by Hossen et al. [6], who proposed the XMB-BERT framework—a hybrid architecture combining embeddings from BanglaBERT, mBERT, and XLM-RoBERTa, followed by dimensionality reduction using PCA and classification through a voting mechanism. Evaluated on a 4,200-sample Bangla social media dataset, their approach achieved an accuracy of 83.7%. This dataset forms the foundation of the current study.

In contrast to prior work that primarily uses transformer embeddings as features for downstream classifiers, this study investigates both classical feature-based models and fine-tuned transformer architectures. By directly comparing these approaches under a consistent experimental setup, this work aims to provide practical insights into the effectiveness of different modeling strategies for Bangla sentiment analysis.

While prior works have primarily focused on improving performance through hybrid architectures and ensemble strategies, relatively fewer studies have explored the effectiveness of simpler baseline models under the same experimental conditions. This gap motivates the current study, which aims to provide a direct and consistent comparison between classical machine learning approaches and fine-tuned transformer models on the same dataset.

III. DATASET DESCRIPTION

The dataset used in this study is the Student-People Mass Uprising Public Sentiments Dataset, originally introduced by Hossen et al. [6]. It consists of 4,200 manually collected Bangla-language social media comments gathered during and after the July Revolution of 2024 from three platforms: Facebook, YouTube, and Twitter (X). The dataset contains four attributes: comment text, language, platform, and sentiment label.

A. Annotation and Class Distribution

Each comment is annotated into one of three sentiment classes: *Positive*, *Negative*, and *Neutral*, with an initially balanced distribution of 1,400 samples per class. During preprocessing, duplicate comments were identified and removed, reducing the dataset to 4,083 unique instances. The final class distribution remained approximately balanced, with 1,375 positive, 1,367 negative, and 1,341 neutral samples. No missing values were present in the dataset.

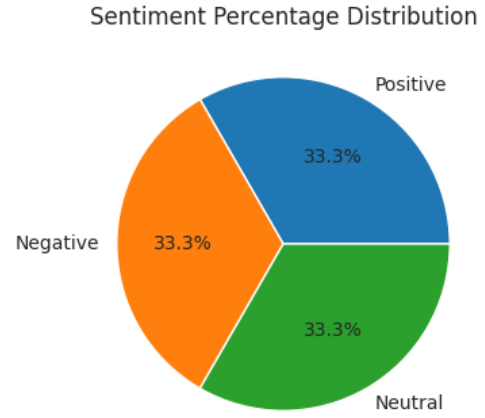


Fig. 1: Sentiment percentage distribution (33.3% per class).

B. Platform Distribution and Statistics

Platform-wise distribution revealed an uneven spread of data, with Facebook contributing the largest portion (1,711 samples), followed by YouTube (1,399) and Twitter (1,090). Notably, sentiment distribution across platforms is highly skewed; for example, Twitter contains very few positive samples compared to other classes, which may introduce bias in model learning.

Text statistics indicate that the dataset consists of short, informal Bangla comments typical of social media, with an average length of approximately 10.8 words and 68.8 characters per comment. Text statistics indicate that the dataset consists of short, informal Bangla comments typical of social media, with an average length of approximately 10.8 words and 68.8 characters per comment. The presence of spelling variations, informal expressions, and inconsistent punctuation contributes to the noisy and unstructured nature of the data, making sentiment classification more challenging.

The observed imbalance in platform-specific sentiment distribution introduces an additional layer of complexity, as models may learn biases associated with particular platforms. This highlights the importance of considering data source variability when interpreting model performance and generalization capability.

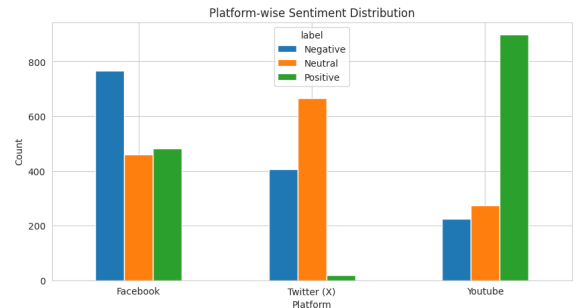


Fig. 2: Platform-wise sentiment distribution across Facebook, Twitter (X), and YouTube.

IV. METHODOLOGY

A. Data Preprocessing

Two distinct preprocessing pipelines were employed based on the model type.

For classical machine learning models, aggressive preprocessing was applied, including normalization, punctuation removal, and stopword filtering using a predefined list of 121 Bangla stopwords. This resulted in cleaner and more compact textual representations suitable for TF-IDF feature extraction.

In contrast, minimal preprocessing was applied for transformer-based models to preserve contextual and syntactic information. The original sentence structure was largely retained, as transformer architectures rely on contextual embeddings derived from raw text.

Duplicate removal was performed prior to dataset splitting to prevent data leakage. The dataset was divided into training and test sets using an 80–20 split for baseline models. For transformer models, an additional validation set was created, resulting in splits of approximately 80% training, 10% validation, and 10% test.

The difference in preprocessing strategies reflects the fundamental differences between feature-based and representation-based models. While classical models benefit from noise reduction and feature sparsity, transformer models rely on contextual relationships within the original text, making minimal preprocessing more suitable.

B. Feature Extraction

For classical models, features were extracted using TF-IDF with a character n-gram representation (`char_wb`, n-gram range 2–5) and a maximum of 10,000 features. This approach captures subword patterns and is particularly effective for morphologically rich languages like Bangla, where spelling variation is common in informal social media text.

The use of character n-grams is particularly effective for Bangla due to its rich morphology and spelling variations. This improves robustness compared to word-level features.

Transformer-based models utilized tokenized input sequences and contextual embeddings generated internally by their pretrained architectures, leveraging wordpiece and sentencepiece tokenizers appropriate to each model.

C. Models

1) *Classical Machine Learning Models*: Three classical machine learning models were implemented using TF-IDF features:

- **Logistic Regression (LR) [10]**: A linear classifier that computes a weighted sum of input features and applies a softmax over the three sentiment classes. Well-suited for high-dimensional sparse TF-IDF vectors.
- **Naive Bayes (NB) [11]**: A probabilistic classifier based on Bayes' theorem with conditional independence assumptions. Known for efficiency on text classification tasks.

- **Random Forest (RF) [12]**: An ensemble of decision trees that aggregates predictions through majority voting, capturing non-linear feature interactions.

2) *Transformer-Based Models*: Three pretrained transformer models were fine-tuned for three-class sequence classification:

- **BanglaBERT [3]**: A BERT-based model pretrained on large-scale Bangla corpora. Its language-specific pretraining is particularly advantageous for Bangla NLP tasks.
- **XLM-RoBERTa [8]**: A multilingual RoBERTa model trained on 100 languages. Provides strong cross-lingual transfer capabilities.
- **mBERT [7]**: Multilingual BERT pretrained on 104 languages. A widely used baseline for multilingual NLP evaluation.

All transformer models were fine-tuned using a classification head with three output labels. Training was conducted for multiple epochs with early stopping to mitigate overfitting.

D. Training Configuration

Classical models were trained using scikit-learn with default hyperparameters, except Logistic Regression, which used `max_iter=1000` and `C=1.0`. Transformer models were fine-tuned using the AdamW optimizer with a learning rate of 2×10^{-5} , batch size of 32, and a maximum sequence length of 128 tokens. Early stopping was applied based on validation loss with patience of 3 epochs.

E. Tools and Libraries

The implementation was carried out using Python. Key libraries include scikit-learn for classical models and evaluation metrics, HuggingFace Transformers for pretrained models and fine-tuning, PyTorch for the deep learning backend, Pandas and NumPy for data processing, and Matplotlib and Seaborn for visualization.

F. Evaluation Metrics

Model performance was evaluated using Accuracy, Precision, Recall, and Macro F1-score, with particular emphasis on Macro F1 due to the multi-class and approximately balanced nature of the task. The metrics are defined as follows, where TP , FP , FN , and TN denote true positives, false positives, false negatives, and true negatives, respectively [13]:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

Macro F1 is computed as the unweighted mean of per-class F1-scores:

$$\text{Macro-F1} = \frac{1}{C} \sum_{c=1}^C \frac{2 \cdot P_c \cdot R_c}{P_c + R_c} \quad (4)$$

where $C = 3$ is the number of sentiment classes, and P_c , R_c denote class-level precision and recall, respectively.

TF-IDF remains particularly effective for short-text classification tasks, as it captures term importance without requiring large-scale training data. In the context of Bangla social media text, where messages are typically concise and informal, character-level representations help mitigate issues arising from inconsistent spelling and morphological variation.

V. RESULTS AND EVALUATION

A. Overall Performance Comparison

Table I summarizes the performance of all models across the test set. BanglaBERT achieved the best overall performance, while Logistic Regression emerged as the strongest classical baseline.

TABLE I: Model Performance on the Test Set

Model	Accuracy	Precision	Recall	Macro F1
BanglaBERT	0.8704	0.8813	0.8705	0.8726
mBERT	0.8484	0.8551	0.8488	0.8500
XLM-RoBERTa	0.8460	0.8479	0.8466	0.8467
Logistic Regression	0.8433	0.8535	0.8436	0.8445
Random Forest	0.8250	0.8335	0.8253	0.8261
Naive Bayes	0.8115	0.8183	0.8121	0.8114

It can be observed that although BanglaBERT achieves the highest performance, the improvement over Logistic Regression is relatively small (approximately 2.8% in Macro F1), indicating that simpler models remain effective for this dataset.

In addition, Random Forest and Naive Bayes demonstrate comparatively lower performance, indicating that ensemble tree-based methods and probabilistic classifiers may be less effective in handling sparse high-dimensional TF-IDF features for this task. The consistent ranking across models further validates the robustness of the experimental setup.

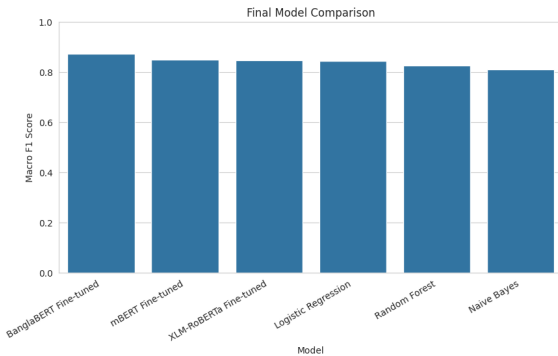


Fig. 3: Final model comparison by Macro F1 score across all six models.

B. Class-wise Performance

The higher accuracy observed for the Neutral class suggests that neutral statements tend to contain less ambiguous sentiment cues compared to Positive and Negative classes. In contrast, expressions of positive and negative sentiment often

involve nuanced language, sarcasm, or contextual dependencies, contributing to increased misclassification rates between these two classes.

Across all models, the Neutral class consistently achieved the highest per-class F1-score, while Positive and Negative classes exhibited relatively lower recall in certain configurations. This may be attributed to subtle linguistic differences and context ambiguity in sentiment expression, particularly for ironic or mixed-sentiment comments common in political discourse.

C. Training Behavior of Transformer Models

Transformer models exhibited signs of overfitting after a few training epochs, as validation loss increased despite stable accuracy. This behavior indicates that model capacity may not be fully utilized given the relatively small dataset size of 4,083 samples. BanglaBERT showed better generalization compared to the multilingual models, likely due to its language-specific pretraining.

This behavior is consistent with the characteristics of deep transformer architectures, which typically require larger datasets to fully generalize. In smaller datasets, these models may quickly memorize training patterns, leading to overfitting without significant gains in generalization performance.

VI. DISCUSSION

The experimental results provide several important insights into model selection for low-resource Bangla sentiment analysis.

A. Classical vs. Transformer Performance

While BanglaBERT achieves the highest performance, the margin of improvement over the TF-IDF + Logistic Regression baseline is relatively small (Macro F1: 0.8726 vs. 0.8445). This suggests that for small-scale datasets, classical feature-based models remain highly competitive when combined with appropriate preprocessing and feature engineering. The character n-gram TF-IDF representation plays a crucial role in capturing morphological patterns and spelling variations, making it particularly effective for informal social media Bangla.

From a practical perspective, classical models offer significant advantages in terms of training time, computational cost, and interpretability. This makes them a viable alternative in scenarios where resources are limited or rapid deployment is required.

B. Language-Specific vs. Multilingual Transformers

Multilingual transformer models such as XLM-RoBERTa and mBERT perform slightly below BanglaBERT, reinforcing the importance of language-specific pretraining for low-resource languages. The performance gap, though modest, highlights that monolingual pretraining data quality and language coverage directly influence downstream task performance.

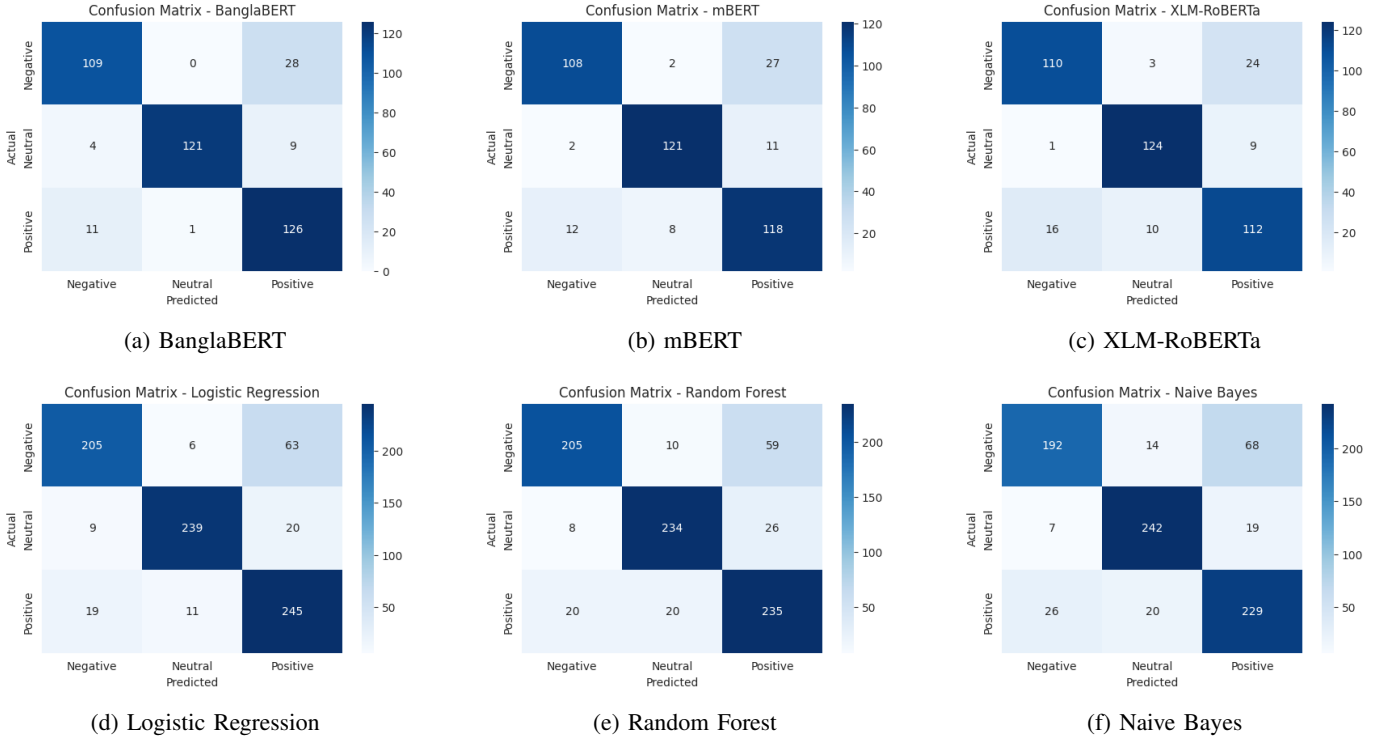


Fig. 4: Confusion matrices for all evaluated models. Transformer-based models (top row) show improved balance across classes compared to classical machine learning models (bottom row). BanglaBERT demonstrates the best overall class-wise performance.

C. Platform Bias

A key observation is the presence of platform-specific bias in the dataset. The extremely low number of positive samples in Twitter data may affect the model’s ability to generalize across platforms. Future work should consider platform-stratified evaluation to better understand cross-platform generalization.

Such platform-specific discrepancies may result in models implicitly associating certain sentiment patterns with specific platforms, thereby reducing their ability to generalize across diverse data sources.

D. Limitations

The dataset size is relatively small, which limits the generalization capability of deep learning models. Additionally, hyperparameter tuning was not extensively explored for transformer models, and more sophisticated architectures or ensemble methods could further improve performance. The absence of cross-validation also limits the statistical reliability of reported metrics.

E. Improvements

Compared to the approach proposed by Hossen et al. [6], several improvements were introduced in this study.

The improvements introduced in this study can be categorized into several methodological and experimental enhancements.

First, transformer models were fine-tuned directly for classification rather than being used only as feature extractors. The prior work relied on extracting embeddings followed by PCA and classical classifiers. In contrast, this study applies end-to-end fine-tuning.

Second, dimensionality reduction was not applied. Avoiding PCA preserves full contextual information.

Third, TF-IDF was improved using character n-grams (2–5), which better capture Bangla morphology.

Finally, duplicate removal was performed prior to train-test splitting, ensuring that no data leakage occurs between training and testing data.

These changes resulted in improved performance, with BanglaBERT achieving a Macro F1-score of 0.8726 compared to approximately 0.837 in prior work.

These improvements collectively contribute to a more robust and reliable evaluation framework, enabling a clearer comparison between classical and transformer-based approaches.

F. Key Findings

Several important observations were made during experimentation.

First, classical models remain competitive. TF-IDF + Logistic Regression achieved a Macro F1-score of 0.8445, close to transformer models.

Second, BanglaBERT outperformed multilingual models, showing the benefit of language-specific pretraining.

Third, the Neutral class achieved the highest accuracy across all models.

Fourth, Positive and Negative classes show higher confusion across all models.

Finally, transformer models showed signs of overfitting after a few epochs.

VII. CONCLUSION

This study presented a comparative analysis of classical machine learning and transformer-based approaches for Bangla sentiment analysis using a social media dataset related to the July Revolution of 2024. Experimental results demonstrate that BanglaBERT achieves the best overall performance (Macro F1: 0.8726), highlighting the effectiveness of language-specific transformer models. However, a well-tuned TF-IDF + Logistic Regression model achieves comparable results (Macro F1: 0.8445), demonstrating that simpler models can be highly effective for small datasets.

The findings emphasize the importance of feature representation, preprocessing strategies, and dataset characteristics in determining model performance. Character n-gram TF-IDF proves particularly valuable for Bangla's morphological richness, and classical models should not be dismissed in favour of deeper architectures when data is scarce.

Future work will focus on improving model generalization through larger and more diverse datasets, advanced fine-tuning strategies such as domain-adaptive pretraining, cross-validation, and hybrid or ensemble architectures combining transformer embeddings with statistical features.

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